Concentrate or Collapse: Exact Browser Success Reveals Concentration Failure in Diffusion Web Agents

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Abstract

Diffusion language models are attractive for browser agents because action plans are structured objects, not purely left-to-right text. We evaluate whether that advantage survives exact browser execution. On FormFactory, we compare an autoregressive control (Qwen3-8B), a masked diffusion model with a causal backbone (ReFusion-8B), and a pure discrete flow matching model (FS-DFM-1.3B) using browser-native metrics: exact completion (**Pass@1**), parser validity, full browser execution, and average reward. On a matched greedy *test* split, ReFusion reaches 4.8% Pass@1 after supervised fine-tuning (SFT), remains at 4.8% under Standard GRPO, improves directionally to 7.3% under ESPO, and reaches 8.9% under MDPO. FS-DFM stays at 0.0% greedy Pass@1 across SFT, AR-surrogate GRPO, ESPO, CJ-GRPO, and MDPO. A recovered autoregressive control reaches 20.2% after SFT and 21.8% after GRPO on the same *test* split, despite 100% parser validity and execution. The resulting picture is not "AR solves the task, diffusion fails." Instead, exact browser success is much harsher than reward for every family. Stochastic decoding further shows that FS-DFM often produces parseable browser plans, but those plans do not concentrate into a usable greedy policy. This makes browser-agent evaluation a measurement problem first, and an RL-design problem second.

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1 Introduction

Browser agents must turn language into executable action sequences: click the correct field, type the correct value, select the correct option, and submit the form in the correct state. This is stricter than text generation alone. A plan can look plausible, earn partial reward, and still fail the real task. For that reason, exact browser completion is a more meaningful endpoint than permissive proxy metrics.

We study this issue for diffusion language models on web form filling. Diffusion policies are appealing because browser plans are structured objects with long-range dependencies across fields, action order, and submission state. In robotics and planning domains, diffusion policies often benefit from modeling entire action objects rather than predicting a single next token at a time (Chi et al., 2023). In language, however, diffusion training and reinforcement learning remain unsettled, especially when the deployed policy must decode greedily (Zhao et al., 2025; Liu et al., 2025; Ou et al., 2025; Yang et al., 2025; He et al., 2025).

The recovered exact-success artifacts change the natural story one might tell from reward alone. Under browser-native **Pass@1**, even the autoregressive control is far from ceiling. ReFusion can complete a small but nonzero fraction of held-out tasks under greedy decoding. FS-DFM, in contrast, collapses to 0.0% greedy Pass@1 across all recovered methods. Yet stochastic decoding reveals high parser validity for several FS-DFM checkpoints. That gap points to a specific failure mode: browser competence exists in the distribution, but it does not *concentrate* into a deterministic policy.

This paper therefore asks two concrete questions. First, can a diffusion policy become a competitive browser agent under deterministic execution? Second, which post-training choices improve sampled browser behavior without actually improving greedy executability? These are browser-agent questions first, and diffusion-RL questions second.

From an STAI-X perspective, this yields three contributions. First, we recast the study as a browser-agent measurement paper: **Pass@1** changes the conclusion relative to reward-only reporting. Second, we provide a browser-native decomposition into exact success, parser validity, and full execution, showing where performance is lost. Third, we identify concentration failure as the main mechanism separating the two diffusion families under greedy decoding.

2 Related Work and STAI-X Positioning

Recent reinforcement learning work for discrete diffusion language models has focused on math, reasoning, and coding benchmarks rather than browser interaction. diffu-GRPO and Flow-GRPO adapt policy-gradient style objectives to diffusion generation (Zhao et al., 2025; Liu et al., 2025). More recent sequence-level methods, including ESPO, CJ-GRPO, and MDPO, argue that the correct optimization target is the quality of the whole generated sequence rather than a collection of tokenwise terms (Ou et al., 2025; Yang et al., 2025; He et al., 2025). Our results agree with that shift in spirit, but they also show that browser-agent conclusions depend strongly on the metric used to evaluate the resulting policy.

This is where the STAI-X angle matters. The paper is not just about another RL objective. It is about how a statistical choice of endpoint changes the scientific conclusion. A shaped reward can be useful for optimization, but it is a noisy proxy for browser completion. Exact browser success, parser validity, full execution, and the greedy-stochastic gap reveal different latent failure modes. That decomposition is much closer to the kind of measurement question STAI-X reviewers are likely to care about than a raw reward leaderboard.

Our work is also distinct from prior browser-agent RL papers such as WebRL (Liu et al., 2025c). Those systems use autoregressive policies and do not ask whether diffusion distributions can be made compatible with greedy deployment. In our setting, that compatibility question is central. We also deliberately scope this version away from an efficiency claim. Diffusion-agent papers often motivate latency or cost advantages, but we do not yet have a reviewer-clean latency/cost measurement stack on the recovered checkpoints, so the STAI-X version focuses on deterministic executability and concentration.

3 Browser-Agent Setting and Metrics

We use FormFactory (Li et al., 2025a), a benchmark of 1,240 web form-filling tasks spanning 25 form types and multiple domains. Each task contains a form URL and ground-truth field values. We use the standard 992/124/124 train/validation/test split.

Each model outputs a textual action plan that is parsed into browser operations and executed in headless Chromium. We report four metrics.

- **Pass@1**: the fraction of prompts whose executed run reaches the success page.
- **Parser validity**: the fraction of prompts that yield at least one parseable browser action.
- **Full execution**: the fraction of prompts whose parsed actions all execute successfully.
- **Average reward**: the mean shaped reward used during online training.

This hierarchy is deliberate. For browser agents, **Pass@1** is the deployment metric. Parser validity and full execution are diagnostic metrics that explain why exact success is lost. Average reward

Table 1: Model families and evaluation role in the STAI-X paper.

Family	Example model	RL methods shown	Paper role
AR control	Qwen3-8B	SFT, GRPO	Measurement anchor
Masked diffusion	ReFusion-8B	SFT, Std. GRPO, ESPO, MDPO	Flagship case
Pure DFM	FS-DFM-1.3B	SFT, AR surrogate, ESPO, CJ-GRPO, MDPO	Stress test for concentration

is supporting evidence only, because it mixes exact completion, field similarity, and execution completeness.

Formally, for task input x, let $p_\theta(a \mid x)$ denote the model's distribution over action sequences a, let $g_\theta(x)$ denote the deterministically decoded sequence, and let $S(x, a)$, $P(x, a)$, and $E(x, a)$ denote indicators for exact success, parser validity, and full execution. We then distinguish

$$\text{Pass@1}_{\text{greedy}} = \mathbb{E}_x[S(x, g_\theta(x))], \tag{1}$$

$$\text{Pass@1}_{\text{sample}} = \mathbb{E}_x \mathbb{E}_{a \sim p_\theta(\cdot|x)}[S(x, a)]. \tag{2}$$

The corresponding *concentration gap* is

$$\Delta_{\text{conc}} = \text{Pass@1}_{\text{sample}} - \text{Pass@1}_{\text{greedy}}, \tag{3}$$

with analogous parser and execution gaps obtained by replacing S with P or E. Positive gaps mean that competence exists in the model distribution but does not concentrate into the deterministically deployed policy. Appendix A gives the exact identity and a short proof sketch showing why this gap is the correct object for diagnosing browser-agent concentration failure.

4 Experimental Design

We evaluate three model families. The autoregressive control is Qwen3-8B, trained with SFT and GRPO. The first diffusion family is ReFusion-8B (Li et al., 2025b), a masked diffusion model built on a causal LM backbone. This is the flagship diffusion case in the current paper because it is the only family that achieves nonzero greedy exact success on the matched test split. The second is FS-DFM-1.3B (Karimi Monsefi et al., 2026), a pure discrete flow matching model without an autoregressive backbone. This serves as the stress test for concentration failure. The architectural contrast matters because Standard GRPO naturally uses autoregressive log probabilities, while FS-DFM requires either surrogate objectives or diffusion-native alternatives.

Our comparison spans three RL styles. **Standard GRPO** uses autoregressive log probabilities where available, plus an FS-DFM AR-surrogate ablation to reduce algorithmic confounding (Shao et al., 2024). **Sequence-level methods** include ESPO (Ou et al., 2025), CJ-GRPO (Yang et al., 2025), and MDPO (He et al., 2025). Historical token-level Flow-GRPO iterations are useful context (Liu et al., 2025), but the clean STAI-X tables focus on recovered exact-success checkpoints with consistent per-prompt browser fields.

The recovered Qwen exact-success artifacts are now available on the same *test* split as the diffusion tables, so the main browser-success comparison is fully matched across families. We still use Qwen primarily as a measurement anchor rather than a leaderboard winner, because exact browser completion remains far from ceiling even for the control.

The online reward used during RL is

$$r = 0.4\, r_{\text{task}} + 0.4\, r_{\text{field}} + 0.2\, r_{\text{exec}}, \tag{4}$$

where r_{task} indicates whether the success page was reached, r_{field} measures field-level similarity, and r_{exec} measures how many planned actions actually executed in the browser. This reward is useful for training because it gives credit for partial progress, but it is also precisely why reward-based reporting can be misleading: a model can fill many fields correctly, execute almost every action, and still fail the browser task.

For the main STAI-X comparisons, we report raw success counts, Wilson 95% confidence intervals for Pass@1, and a small number of Fisher exact tests for the headline pairwise questions. We

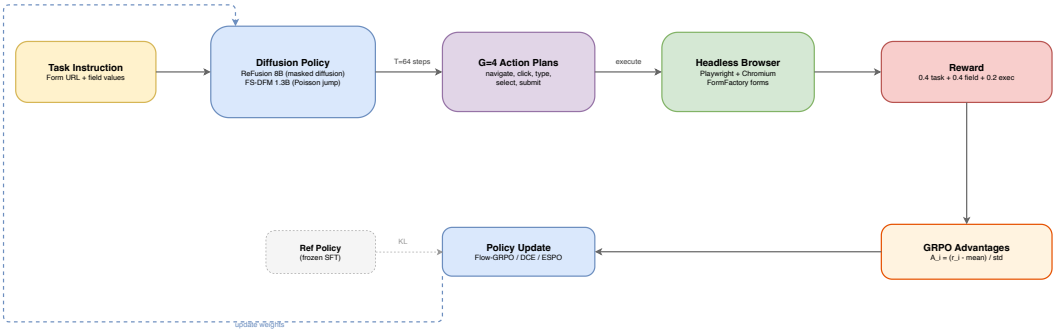


Figure 1: Online browser-agent training and evaluation pipeline. A policy generates action plans, the plans are executed in a real browser, and the resulting traces yield both exact browser success and finer-grained diagnostics.

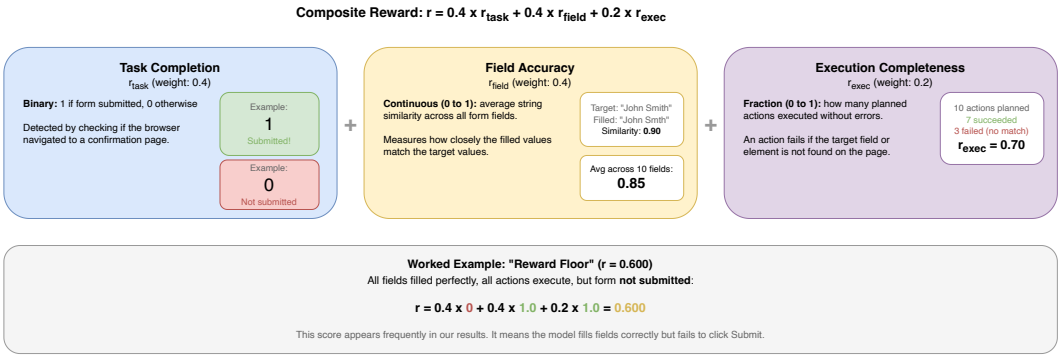


Figure 2: Why reward and exact success diverge. A run can earn substantial reward through correct field values and mostly successful action execution while still failing to reach the success page.

intentionally do not turn this into a large significance-testing exercise. With only 124 prompts per split, the most useful statistical service here is to keep the interpretation honest, not to manufacture a ranking from noisy differences. This paper is strongest when read as a measurement-and-mechanism paper, not as a final leaderboard.

5 Results

5.1 Exact Browser Success Is Much Harsher Than Reward

Table 2 is the main result. The most important observation is not which method "wins," but how low exact browser success remains once reward proxies are removed from the headline. Qwen reaches only 20.2% after SFT and 21.8% after GRPO on the matched *test* split, despite 100% parser validity and full execution. For the diffusion families on *test*, ReFusion is the only family with nonzero greedy Pass@1. FS-DFM remains at 0.0% across all recovered greedy checkpoints.

This table rules out two easy narratives. The first is that exact browser success is essentially solved by autoregressive models. It is not. The second is that all diffusion models fail equally. They do not. ReFusion achieves modest greedy task completion, while FS-DFM fails much earlier, often before it produces any parseable plan at all.

It is equally important that the average-reward view would tell a much softer story. Qwen's average reward is above 0.63 in both recovered control runs, yet exact completion stays below 22%. ReFusion SFT averages 0.433 reward on the test split while completing only 6 of 124 tasks exactly. The statistical lesson is that reward is a noisy surrogate for the event of interest. In this benchmark, partial field correctness and near-complete execution are common enough that they can mask poor end-to-end completion.

Table 2: Browser-native greedy evaluation on the matched *test* split for all model families.

Model	Method	Split	Pass@1	Parser	Exec
Qwen3-8B	SFT	test	25/124 (20.2%)	100.0%	100.0%
Qwen3-8B	GRPO	test	27/124 (21.8%)	100.0%	100.0%
ReFusion-8B	SFT	test	6/124 (4.8%)	91.1%	83.1%
ReFusion-8B	Standard GRPO	test	6/124 (4.8%)	89.5%	82.3%
ReFusion-8B	ESPO μ=8	test	9/124 (7.3%)	83.1%	71.8%
ReFusion-8B	MDPO	test	11/124 (8.9%)	91.9%	82.3%
FS-DFM-1.3B	SFT	test	0/124 (0.0%)	0.0%	0.0%
FS-DFM-1.3B	AR surrogate	test	0/124 (0.0%)	0.0%	0.0%
FS-DFM-1.3B	ESPO μ=8	test	0/124 (0.0%)	0.0%	0.0%
FS-DFM-1.3B	CJ-GRPO	test	0/124 (0.0%)	0.0%	0.0%
FS-DFM-1.3B	MDPO	test	0/124 (0.0%)	0.0%	0.0%

Table 3: Small-sample statistical framing for headline Pass@1 comparisons on the matched *test* split. These tests are intentionally few and focused.

Comparison	Wilson 95% CI for higher row	Fisher p
ReFusion SFT vs Std. GRPO	2.2% to 10.2%	1.000
ReFusion SFT vs ESPO μ=8	3.9% to 13.2%	0.596
ReFusion SFT vs MDPO	5.0% to 15.2%	0.315
Qwen3 SFT vs Qwen3 GRPO	15.4% to 29.8%	0.876

5.2 Why Reward Overstates Browser-Agent Competence

Figure 2 illustrates the source of the mismatch. The reward assigns 40% weight to exact task completion, but the remaining 60% comes from field similarity and execution completeness. That is a reasonable engineering choice for RL, because a sparse binary reward alone would be difficult to optimize. It is a poor choice for the headline scientific claim, because the reward mixes several different concepts of success.

This distinction is especially visible for the autoregressive control. Qwen parses and executes essentially every plan, which means it does well on the reward components tied to syntax and browser interaction. But those strengths are not enough to produce exact task completion on most prompts. The same logic applies to ReFusion. It is already much more browser-compatible than FS-DFM under greedy decoding, yet its exact completion remains sparse. In short, reward is diagnosing *partial competence*, while Pass@1 is diagnosing *usable completion*.

5.3 The ReFusion Gains Are Directional, Not Decisive

The recovered sample size is only 124 tasks per split, so the correct interpretation must be cautious. Wilson 95% confidence intervals for ReFusion are 2.2% to 10.2% for SFT, 3.9% to 13.2% for ESPO μ=8, and 5.0% to 15.2% for MDPO. Two-sided Fisher exact tests do not establish a statistically significant Pass@1 difference between SFT and any recovered ReFusion sequence-level checkpoint at this sample size: SFT vs ESPO gives $p = 0.596$, and SFT vs MDPO gives $p = 0.315$.

This matters for the paper's framing. The data supports a *directional* claim that sequence-level methods are more promising than Standard GRPO for ReFusion under exact browser success. It does *not* support a strong ranking claim such as "MDPO definitively beats ESPO" or "ReFusion has been solved by RL." In the STAI-X version, the cleaner contribution is the evaluation correction itself.

Table 3 should be read as a guardrail on over-interpretation, not as a failed analysis. The key message is that the recovered ReFusion improvements are consistent with a positive sequence-level trend, but the sample is too small to justify a strong ranking claim among SFT, ESPO, and MDPO. The same logic applies to the Qwen control: GRPO looks directionally better than SFT, but the evidence is not precise enough for a strong claim. That is exactly why the paper emphasizes measurement correction and concentration diagnostics over winner-take-all method ranking.

Table 4: Greedy versus stochastic decoding on the test split. Each cell shows Pass@1 / parser validity. This table operationalizes the greedy-stochastic execution gap proposed in the project plan, and the parser-validity gap is the clearest concentration diagnostic.

Checkpoint	Greedy	Stochastic
FS-DFM SFT	0.0 / 0.0	0.0 / 84.7
FS-DFM AR surrogate	0.0 / 0.0	0.8 / 87.1
FS-DFM ESPO μ=8	0.0 / 0.0	1.6 / 87.9
FS-DFM CJ-GRPO	0.0 / 0.0	0.8 / 83.9
FS-DFM MDPO	0.0 / 0.0	0.8 / 76.6
ReFusion SFT	4.8 / 91.1	5.6 / 85.5
ReFusion Standard GRPO	4.8 / 89.5	4.8 / 90.3
ReFusion ESPO μ=8	7.3 / 83.1	7.3 / 84.7
ReFusion MDPO	8.9 / 91.9	5.6 / 83.1

5.4 FS-DFM Learns Partial Browser Competence but Fails to Concentrate

The most informative diagnostic is the greedy-stochastic gap in Table 4. For FS-DFM, the exact-success gaps are numerically small, but the parser-validity gaps are large. Greedy decoding produces no parseable plans for the selected checkpoints. Stochastic decoding, however, often yields parser validity between 76.6% and 87.9% on the same test prompts. That is not random noise. It indicates that the model has learned a nontrivial distribution over browser-like plans, but greedy decoding does not recover a consistent mode.

ReFusion looks different. Its stochastic and greedy parser-validity values stay close, and its exact-success gaps are small or even negative. That pattern is consistent with a more concentrated policy family. The implication is architectural as much as algorithmic. ReFusion's causal backbone appears to make greedy decoding compatible with the learned distribution, while the pure DiT family remains broad and multimodal.

5.5 Operational Meaning of Concentration Failure

For a deployed browser agent, concentration failure is not an academic detail. Most practical systems use deterministic or near-deterministic decoding for reliability, reproducibility, and cost control. A model that only looks competent under stochastic decoding is therefore not yet an effective browser agent, even if its sampled trajectories contain useful structure.

This is why parser validity is so valuable in the STAI-X framing. It separates "the model never learned browser syntax" from "the model learned a diffuse set of local browser moves but cannot compose them into a stable deterministic plan." FS-DFM is mostly the second case, not the first. That is a much more precise scientific conclusion than simply calling the model collapsed.

Historical token-level RL runs remain useful context for why this version emphasizes sequence-level methods and browser-native exact-success metrics, but they are exploratory rather than submission-grade evidence for the present STAI-X claim. We therefore move that history to the appendix and keep the main body focused on the recovered exact-success artifacts.

5.6 Why the Greedy-Stochastic Gap Is the Right Browser Diagnostic

The concentration view clarifies several otherwise confusing patterns in Table 4. A model can have high stochastic parser validity for two very different reasons. One possibility is that it has learned the benchmark well and simply needs a better deterministic decoder. Another is that it has learned a broad cloud of locally plausible traces that share browser syntax but not the task semantics needed for exact completion. The diagnostic value of the table comes from separating those cases.

FS-DFM is the strongest example. Under stochastic decoding, several checkpoints produce parseable plans on more than four out of five prompts. That means the model has internalized a substantial amount of browser grammar and local action structure. However, its sampled exact success remains close to zero, and its greedy policy never even reaches parseability. The scientific conclusion is therefore not just that the greedy decoder is poor. It is that the learned distribution is diffuse in

precisely the wrong way for browser deployment: mass exists on browser-like plans, but not on a stable, task-completing mode.

ReFusion presents the opposite profile. Its stochastic and greedy parser-validity numbers are similar, and its exact-success rates stay in the same range across decoding modes. That does not mean ReFusion is already a strong browser agent. It means the failure is mostly *coverage-limited* rather than *concentration-limited*. In other words, when ReFusion knows how to do a task, greedy decoding is not the main bottleneck. For FS-DFM, greedy decoding *is* the bottleneck.

5.7 Measurement Choice Changes the Scientific Claim

This benchmark is a useful reminder that metric choice is not a cosmetic reporting decision. If we headline average reward or nonzero reward rate, then the story becomes that all three model families show meaningful browser progress, with FS-DFM merely lagging behind. If we headline exact browser success, then the story changes sharply: the autoregressive control is still far from reliable, ReFusion is only directionally improved by RL, and FS-DFM fails to produce any deterministic task completion at all on the recovered test checkpoints.

From a statistical perspective, this is exactly the kind of endpoint issue that should matter at STAI-X. The reward combines a binary event of substantive interest with two auxiliary quantities that are easier to optimize. That mixture is useful for policy learning, but it is not an unbiased estimator of browser completion, nor is it calibrated to deployment risk. In the present setting, two agents with similar average reward can differ materially in exact task completion because one is better at partial field matching while the other is better at actually closing the task.

The practical implication is that browser-agent papers should report a metric stack rather than a single scalar. Pass@1 answers whether the agent solves the task. Parser validity answers whether it speaks the browser's action language. Full execution answers whether the plan survives contact with the environment. The greedy-stochastic gap answers whether the learned competence has concentrated into a usable deterministic policy. Each metric corresponds to a different scientific failure mode, and collapsing them into one reward number hides those distinctions. A visual summary of this concentration logic is included in the appendix.

6 Discussion

From a reviewer perspective, the safest interpretation is a three-part one.

First, browser-agent evaluation should lead with exact completion. If the paper had stayed with nonzero reward as its headline metric, it would have overstated both the autoregressive control and the diffusion models. Exact browser execution is a harsher metric, but it is the one that matters for deployment.

Second, ReFusion is a viable but limited browser planner. It reaches nonzero greedy Pass@1 under SFT and remains the only evaluated diffusion family to do so on the matched test split. Sequence-level RL helps directionally, but the gains are small enough that the paper should not oversell them.

Third, FS-DFM should not be summarized as "unable to learn." That would ignore the stochastic results. A better reading is that it learns browser-like local structure but fails to concentrate that structure into a deterministic policy. For browser agents, that distinction matters because deployment is usually greedy or near-greedy.

This interpretation also clarifies how the current paper should discuss work not yet run. Additional benchmarks such as REAL or WebArena, and additional diffusion baselines such as Dream or a stabilized LLaDA pipeline, are important next tests. Latency and cost are also important, especially if diffusion is to compete with AR on efficiency rather than only on capability. But those are extensions of the present claim, not prerequisites for the current one. The current evidence is already sufficient for a focused FormFactory result about exact browser success and concentration.

7 Threats to Validity and Planned Extensions

This version has five important limitations. The exact-success evaluation sets are small ($n = 124$ each), which weakens pairwise significance claims. We do not yet include broader browser benchmarks such as WebArena or REAL. We also do not include newer diffusion baselines such as Dream or a cleaned LLaDA run. We report one recovered artifact per checkpoint rather than a multi-seed campaign. Finally, we do not yet provide reviewer-clean latency or cost-per-task measurements, so this paper does not claim an efficiency advantage for diffusion browser agents.

A second threat is architectural confounding. ReFusion is both larger and initialized from a causal backbone, while FS-DFM is smaller and trained as a pure diffusion transformer. The AR-surrogate ablation reduces one part of that confound by showing that simply giving FS-DFM a more AR-like objective does not restore greedy exact success. It does not fully remove the scale difference. A stronger future design would include a smaller causal control or a larger pure-DiT control.

A third threat is benchmark scope and variance. FormFactory is a real browser benchmark, but it is still a form-filling benchmark rather than a full long-horizon web benchmark, and the present paper summarizes one recovered artifact per checkpoint rather than a multi-seed campaign. That is acceptable for a focused STAI-X paper when paired with exact counts and uncertainty intervals, but broader benchmarks and multi-seed reruns remain the right next step.

7.1 Practical Evaluation Implications

The present results suggest a simple reporting discipline for browser-agent papers. First, exact browser completion should be the headline endpoint, because it aligns with the real deployment event of interest. Second, parser validity and full execution should be reported alongside it, because they localize whether failures arise from syntax, environment interaction, or true task-level reasoning. Third, when a model family supports both sampled and deterministic decoding, the greedy-stochastic gap should be treated as a first-class diagnostic rather than an afterthought.

This reporting stack matters beyond the specific diffusion architectures studied here. A browser agent can look promising under reward, sampled rollouts, or partial field metrics while still failing the operational question a user actually cares about: does the deterministic run finish the task? In that sense, the evaluation lesson is broader than the method comparison. The paper's main scientific claim is not only that FS-DFM is concentration-limited on FormFactory, but also that browser-agent measurement must expose that failure mode explicitly.

For STAI-X specifically, this is also a statistical design point. The target estimand should match the scientific claim. If a paper claims an agent can complete browser tasks reliably, then the primary estimand should be exact completion under the actual deployment decoder, not a looser mixture of partial-success proxies. That perspective helps explain why the present study is strongest as a measurement-and-mechanism paper: it identifies the endpoint that matters, shows how other metrics drift away from it, and uses the resulting gaps to diagnose where browser-agent competence is or is not concentrated.

8 Conclusion

The main lesson of this STAI-X version is simple: exact browser success changes the story. Under Pass@1, browser planning is harder than reward-only reporting suggests for every model family. ReFusion achieves modest but real greedy completion. FS-DFM does not, despite substantial stochastic parser validity. That makes concentration failure the central mechanism to explain, and browser-native exact success the central metric to report.

The broader implication is methodological. Browser-agent papers should be judged not only by whether competence appears somewhere in the model distribution, but by whether that competence concentrates into a deterministic policy that actually solves tasks. For this version of the paper, that is the strongest and cleanest claim the data supports.

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A Formal Concentration Identity

For each prompt x, define the successful set of action sequences

$$\mathcal{A}^+(x) = \{a : S(x, a) = 1\},$$

where $S(x, a)$ indicates whether execution of a reaches the benchmark success page. Let the deterministic deployment policy be $g_\theta(x)$ and the sampled policy be $a \sim p_\theta(\cdot \mid x)$. Then the prompt-level concentration gap is

$$\Delta_{\text{conc}}(x) = \mathbb{E}_{a \sim p_\theta(\cdot|x)}[S(x, a)] - S(x, g_\theta(x)).$$

Proposition 1. For any prompt x,

$$\Delta_{\text{conc}}(x) = \sum_{a \in \mathcal{A}^+(x)} p_\theta(a \mid x) - \mathbf{1}\{g_\theta(x) \in \mathcal{A}^+(x)\}.$$

Consequently, $\Delta_{\text{conc}}(x) > 0$ whenever the greedy decoded plan fails but the model still assigns positive mass to at least one successful action sequence.

Proof. Because $S(x, a)$ is binary, $\mathbb{E}_{a \sim p_\theta}[S(x, a)] = \sum_a p_\theta(a \mid x) S(x, a) = \sum_{a \in \mathcal{A}^+(x)} p_\theta(a \mid x)$. The greedy term is exactly the indicator that the deterministically chosen sequence is in the successful set. Subtracting the two expressions gives the identity. The final claim follows immediately. □

The same identity applies to parser validity and full execution by replacing S with P or E. This is useful because it separates two kinds of failure. A model may fail because it places little mass on any browser-compatible behavior at all, or it may fail because browser-compatible behavior exists in the distribution but is too diffuse for the deterministic decoder to recover. The FS-DFM results in the main paper are consistent with the second case at the parser-validity level and with both cases at the exact-success level.

B Why Reward Cannot Serve as the Headline Endpoint

The online reward used during training is

$$r = 0.4\, r_{\text{task}} + 0.4\, r_{\text{field}} + 0.2\, r_{\text{exec}}.$$

Taking expectations over prompts and sampled trajectories yields

$$\mathbb{E}[r] = 0.4\, \Pr(r_{\text{task}} = 1) + 0.4\, \mathbb{E}[r_{\text{field}}] + 0.2\, \mathbb{E}[r_{\text{exec}}].$$

Since r_{task} is exact completion while r_{field} and r_{exec} reward partial progress, $\mathbb{E}[r]$ does not identify exact browser success without additional assumptions linking the components. In particular, an agent may increase expected reward by improving partial field matching or execution completeness without improving the binary event that matters for deployment.

This explains the empirical mismatch in the recovered artifacts. Qwen and ReFusion both achieve moderate reward while exact completion remains sparse. The reward is therefore appropriate as a training signal but insufficient as a standalone scientific endpoint. In a browser-agent paper, it should support the mechanism story rather than replace it.

C Supplementary Statistical Notes

The main paper reports raw success counts, Wilson intervals, and a small number of Fisher exact tests. That choice is intentional. With only 124 prompts per split, asymptotic claims can easily look stronger than they are. Reporting exact counts anchors the interpretation in events that readers can reason about directly, while Wilson intervals give a conservative sense of uncertainty around sparse proportions.

The same caution applies to cross-family comparison even after the matched test-split recovery. Qwen is now available on the same split as the diffusion checkpoints, but the paper still reports one recovered artifact per checkpoint rather than a multi-seed campaign. That is sufficient for the present qualitative claim that exact browser success is harsh even for the autoregressive control, but it is not sufficient for a strong final AR-versus-diffusion leaderboard claim.

D What a Stronger Follow-Up Paper Would Add

The present STAI-X submission is strongest as a focused paper about exact measurement and concentration on FormFactory. A longer follow-up paper should add four ingredients.

- Multi-seed reruns for the matched autoregressive and diffusion headline rows.
- Additional browser benchmarks such as WebArena and REAL to test whether the concentration pattern is specific to form filling or a broader property of diffusion agents.
- Additional diffusion baselines, especially newer architectures such as Dream or a stabilized LLaDA-style pipeline, to separate architecture effects from objective effects.
- Reviewer-clean latency and cost-per-task measurements, so any efficiency discussion is empirical rather than speculative.

Those additions are substantial, but they are not required for the present paper's central conclusion. The exact-success evaluation already supports a clean statistical point: for browser agents, the decisive question is not only whether competence exists in the model distribution, but whether that competence concentrates into a deterministic policy that actually solves tasks.

E Historical Token-Level Context

The broader project included many earlier token-level RL runs, especially Flow-GRPO-style experiments motivated by diffusion-native likelihoods (Liu et al., 2025). Those runs remain scientifically useful because they explain why the present paper emphasizes sequence-level methods and browser-native evaluation. Across those historical runs, token-level objectives were fragile, highly sensitive to optimization details, and much easier to interpret through reward than through exact browser completion. In other words, they generated intuition, but not the cleanest STAI-X result.

That is why the main body does not present a long token-level leaderboard. The exact-success artifacts that survived cleanly are strongest for the SFT baselines, Standard GRPO, and the recovered sequence-level checkpoints. Rather than forcing incomplete historical artifacts into the main table, the STAI-X version uses token-level history for framing: it motivates the move from tokenwise objectives toward sequence-level objectives and concentration diagnostics, but it does not carry the central empirical claim.

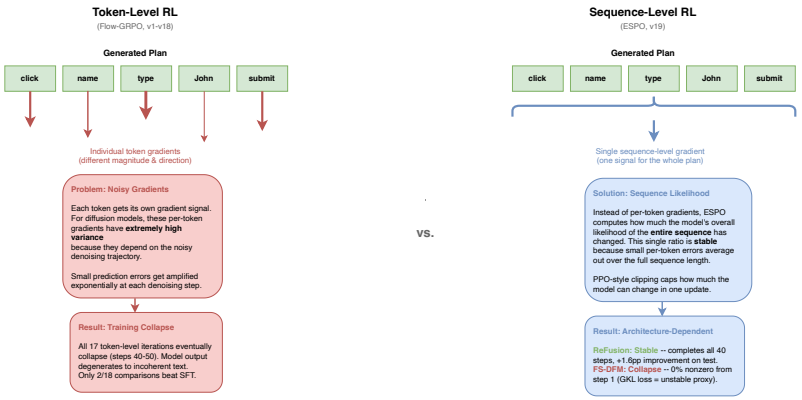


Figure 3: Token-level versus sequence-level optimization. In this paper, the key browser-agent question is whether learned competence concentrates into a usable greedy policy. Sequence-level methods help ReFusion directionally, but the main qualitative distinction comes from the concentration behavior in Table 4.

F Concentration Diagram

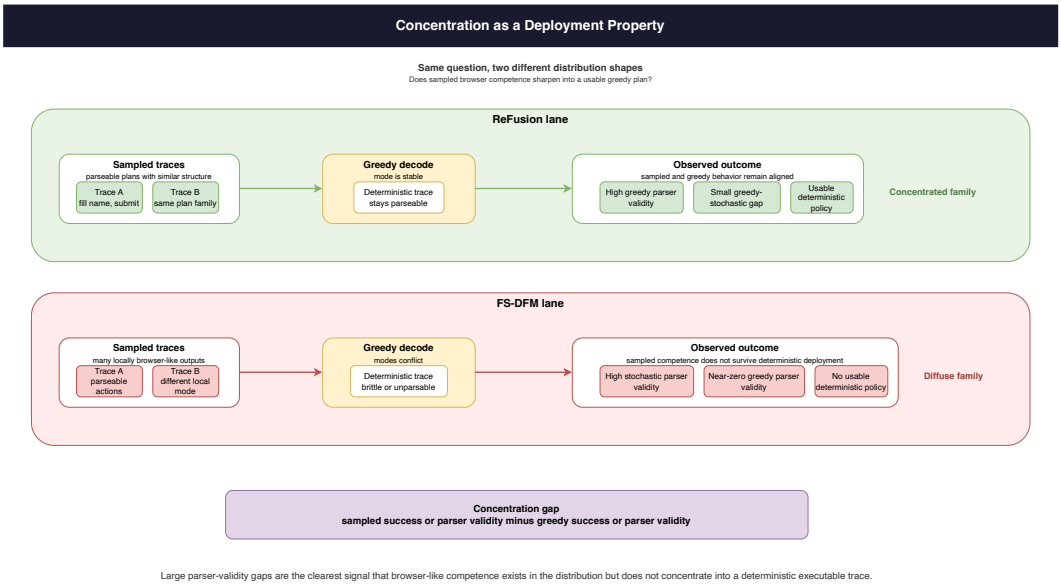


Figure 4: Concentration as a deployment property rather than a sampling property. FS-DFM often places probability mass on locally browser-like traces, but that mass does not sharpen into a reliable greedy action sequence. ReFusion is still weak in absolute terms, but its sampled and greedy behavior are much more aligned.